INCH-POUND
MIL-DTL-24643/72B
27 November 2023
SUPERSEDING
MIL-DTL-24643/72A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, FLEXIBLE, FOR INBOARD USE ONLY (NOT FOR OUTBOARD USE), TYPE LSFCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

Components 1-4:

First	-	Copper conductor, tin-coated, 4 AWG, class K (420 strands of 30 AWG), in accordance with ASTM B172.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white, 0.380±0.010-inch diameter.
Fourth	-	Insulation covering – thermoset insulation, black, 0.450±0.010-inch diameter. Standard identification code applied by method 1, using 1-BLACK through 4-GREEN.

Components 5-8:

First	-	Copper conductor, tin-coated, 16 AWG, ASTM B286 (19 strands of 29 AWG).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, black, 0.090±0.003 inch diameter. Standard identification code applied by method 1, using 5-ORANGE through 8-RED-BLACK.

Distribution Statement A. Approved for public release: distribution is unlimited.

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Assembly:

First - Components 1-4 and 5-8 shall be cabled together with a maximum lay of 11 inches and as follows:

Component 5 shall be placed in the outer interstices between component 1 and 2.

Component 6 shall be placed in the outer interstices between component 2 and 3.

Component 7 shall be placed in the outer interstices between component 3 and 4.

Component 8 shall be placed in the outer interstices between component 4 and 1.

Fillers may be used as necessary to form a firm, well-rounded assembly.

Second - An optional binder.

Third - Braided shield of 34 AWG tin-coated copper. Braid angle of 40 to 50 degrees. Minimum coverage of 90 percent.

Fourth - An optional binder.

Fifth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.075 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL24643).

TABLE I. Details.

Military part no. M24643/72	Type and size	Minimum bend radius (inches)	Overall diameter (inches)	
			min.	max.
-01UO	LSFCF-42	7.5	1.250	1.350

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	0.298
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor (between 4 AWG conductors)	3,000
Conductor to conductor (between 16 AWG conductors)	3,000
Conductor to conductor (between 4 AWG and 16 AWG conductors)	3,000
All conductor to shield	3,000
Insulation resistance (megohms/1,000 feet, min.)	See table I

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Conductor to conductor (between 4 AWG conductors)	500
Conductor to conductor (between 16 AWG conductors)	500
Conductor to conductor (between 4 AWG and 16 AWG conductors)	500
All conductor to shield	250
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Abrasion resistance test	Not applicable
Crack resistance test	Not applicable
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Insulation covering (4 AWG only)	50
Jacket	50
Cold bending (cable) (mandrel diameter 1.350±0.065 times cable diameter)	No failure
Bending endurance (77±3.6 °F [25±2 °C], cycles, min.)	25
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	250
Insulation covering (4 AWG only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Jacket (cable)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Durometer – type A (hardness, min./max.)	60/75
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Insulation covering (4 AWG only)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (overall cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (overall jacket) (cycles, min.)	125
Shrinkage	No failure
Shield – conformance to material (construction and coverage)	No failure
Surface transfer impedance	Not applicable
Shield effectiveness (Electromagnetic pulse [EMP] response)	Not applicable

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Insulation covering (4 AWG only)	2
Jacket	2
Fillers	2
Insulation	2
Durometer – type A (hardness, min./max.)	

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Extruded insulation	60/75
Extruded insulation covering (4 AWG only)	60/75
Extruded jacket	60/75
Halogen content (percent, max.)	
Insulation	0.2
Jacket	0.2
Fillers	0.2
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Electrical moisture absorption	No failure
Weathering (jacket)	Not applicable

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSFCF-42	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-088)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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